Early Detection of Student Burnout & Dropout Risk

Problem Understanding

Academic burnout and dropout happen gradually, but universities usually detect them only after the damage is visible—such as failed courses, repeated absenteeism, or missed submissions. By the time professors notice these warning signs, the student may already be mentally exhausted or disconnected from the learning environment.

Modern learning environments record rich behavioural data through Learning Management Systems (LMS), attendance logs, and feedback platforms. These behavioural patterns—including activity level, consistency, sentiment, and time disorder—act as early indicators of mental fatigue, academic disengagement, and dropout risk.

Objective To build a behavioural analytics system capable of predicting:

- Burnout Risk Level → Low / Medium / High
- Academic Disengagement Indicators
- Dropout Probability (0–1 or %)
- Key Behavioural Triggers
- Intervention Recommendations
- This supports counsellors, faculty, and institutions in taking proactive action, helping students before performance drops.

Data Assumptions / Dataset Description

Real LMS behaviour datasets are usually private. Therefore, we assume synthetic behavioural data, generated to mimic actual student habits.

The dataset contains 10,000 records, each representing one student with weekly behavioural aggregates.

Dataset Structure

A. Behavioural Activity Features

Feature	Description
lms_logins_per_week	Number of LMS logins (1–80)
avg_submission_delay_days	Average delay in assignment submission
attendance_percentage	Student's attendance (40%–100%)
inactivity_gap_days	Longest continuous inactive streak
night_activity_ratio	% of activity between 12 AM–4 AM
study_time_variance	Irregularity in daily study pattern

B. Psychological / Sentiment Features

Feature	Description
sentiment_score	-1 (very negative) to +1 (very positive)
feedback_negativity_flag	Derived from textual feedback

C. Academic Performance Features

Feature	Description
past_sem_gpa	Historical academic performance
gpa_drop_rate	Rate of decline from previous semester

D. Labels

Label	Meaning
burnout_level	Low/Medium/High
dropout_probability	0–1 continuous
disengagement_flag	0/1 (optional target for clustering)

Dataset Behavioural Rules (Simulated Logic)

To generate realistic patterns:

- Low LMS activity + high delay → burnout ↑
- Low attendance (<60%) → dropout ↑
- Negative sentiment → emotional fatigue indicator
- High night activity → stress / exhaustion
- High inactivity gaps → disengagement
- GPA decline → strong dropout predictor

This ensures the dataset closely matches real student behaviour patterns.

Feature Engineering Logic

Raw data does not directly capture burnout. We extract behavioural features that combine multiple signals.

- Engagement Index :
  Represents learning involvement:

engagement_index = 0.5*(normalized LMS logins) + 0.5*(1 - normalized submission delay)

  Higher → more engaged.

- Stress Behaviour Score :
  Combines lifestyle irregularity + night activity:

stress_score = night_activity_ratio × study_time_variance

Higher → more erratic behaviour → early burnout sign.

- Attendance Stability Score :

  attendance_stability = moving_std(attendance_percentage over 4 weeks)

  Higher → inconsistent attendance → risk signal.

- Sentiment Normalization :
  Converts sentiment from -1 to +1 → 0 to 1 scale.

- Risk Behaviour Index (Composite Score):
  Final risk score combining all behavioural signals:

$$\text{risk_behavior_index} = (0.35 \times (1 - \text{engagement_index})) + (0.25 \times \text{stress_score}) + (0.20 \times (1 - \text{sentiment_normalized})) + (0.20 \times (1 - \text{attendance_percentage}))$$

  This index becomes an important input feature to the burnout and dropout prediction models.

Model Selection and Reasoning

Because the system predicts multiple outcomes, we use different ML models for each task.

- Burnout Prediction (Low / Medium / High)

  Model: Random Forest Classifier

  Reason:
  i. Handles nonlinear behavioural patterns
  ii. Works well with categorical burnout levels
  iii. Produces feature importance → helps interpret behaviour
  iv. Robust to noisy data (common in student logs)

- Dropout Probability (0–1 Score)

  Model: XGBoost Regressor

  Reason:
  i. Extremely effective in structured datasets
  ii. Handles missing values internally
  iii. Captures subtle interactions between behavioural + academic features
  iv. Provides smooth probability outputs

- Academic Disengagement Detection

  Model: K-Means Clustering

  To identify naturally occurring behavioural groups:

  i. Highly Engaged
  ii. Moderately Engaged
  iii. At-Risk

  Reason:

  i. Behaviour clustering helps detect patterns before burnout happens
  ii. No label needed
  iii. Helps visually identify risky groups in dashboard

Evaluation Metrics

Because you have different prediction types:

- For Burnout Prediction (Classification)

Metric	Why
Accuracy	Overall correctness
F1-score (macro)	Balances uneven classes
Precision	Avoid false positives
Recall	Avoid missing high-risk students
Confusion Matrix	Shows misclassification patterns

- For Dropout Probability (Regression)

Metric	Why
MAE	Measures average error
RMSE	Penalizes higher errors
R^2 Score	Measures model fit

- For Clustering (Unsupervised)

Metric	Why
Silhouette Score	Cluster separation quality
Davies–Bouldin Index	Compactness check

Behavioural Insights Derived

These insights are extracted from model training + feature importance.

1.Key Indicators of Burnout Based on model weights and correlations:

- High submission delays (most powerful indicator)
- Dropping LMS logins over 2–3 weeks

- Negative sentiment in feedback
- Heavy night-time activity (after 12 AM)
- Erratic study schedule (high variance)

2.Key Indicators of Disengagement Increasing inactivity gaps:

- Missing minor assignments regularly
- Sudden reduction in attendance
- No recent forum participation or feedback submission

3.Dropout Risk Patterns Students with:

- GPA drop > 20%
- Attendance < 60%
- Engagement Index < 0.40
- High Stress Score (>0.7)

show the highest dropout probability.

4.Feature Importance (Example Ranking) attendance_percentage:

- submission_delay
- sentiment_score
- LMS logins
- stress_score
- inactivity_gap

This helps faculty understand what drives risk.

5.Recommended Interventions

- High Risk Counselling session
    a. Reduced coursework load
    b. Weekly mentor check-ins
    c. Mental health support
- Medium Risk Weekly review
    a. Reminders + productivity nudges
    b. Consistency-building study planner
- Low Risk Positive motivation nudges
    a. Encourage balanced routines